

AI-Assisted Survey Question Harmonization: Findings from CPS and FoodAPS Analysis

DRAFT — not validated — see disclaimer slide

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Disclaimer

 Warning

DRAFT — These materials are in development and have not been independently validated.

The views expressed are the author's own and do not necessarily represent the views of the U.S. Census Bureau or the U.S. Department of Commerce.

This is exploratory research assessing feasibility of AI-assisted survey analysis methods.

For detailed methodology validation, see Report 3B: AI Methodology Validation

Why It Matters

For Respondents: - Declining survey response rates + increasing costs - Overlapping questions create unnecessary burden - Consolidation reduces redundant questions

For Federal Agencies: - **Expert time is the scarce resource**, not expert capability - Traditional approach: weeks/months of manual comparison - **Our goal:** Multiply expert effectiveness → faster burden reduction

Dual value proposition: (1) Burden reduction for respondents (2) Expert efficiency to achieve it faster

The Challenge

- Determining consolidation **requires expert judgment** (this doesn't change)
 - Traditional timeline: **weeks to months** for manual review of 1,598 pairs
 - Our approach: **AI handles pre-work, experts focus on highest-value decisions**
 - Result: Structured expert review table ready for validation
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What Experts Get

What AI Provides	What Experts Do
Pre-screened pairs with best matches	Validate match quality
Initial F1/F2/F3 classification	Confirm or override
Documented reasoning	Understand and critique
Confidence scores	Prioritize review effort
Barrier codes	Verify diagnosis

76% classified with high confidence → spot-check **24%** flagged for expert review → where judgment adds most value

Starting Point

- **7,400+** survey questions across **49** federal surveys
 - Mapped to Census Bureau standard taxonomy
 - [Report 1 established concept classification]
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Method Overview

- Multi-model LLM ensemble approach (brief validation — see Report 3B for details)
- Three frontier models to reduce bias
- **High inter-rater agreement (= 0.845)** = well-defined task
- **< 1 day processing time** for work that traditionally takes weeks/months

For detailed methodology validation including construct validity analysis, arbitrator behavioral differences, and cost/quality tradeoffs, see Report 3B

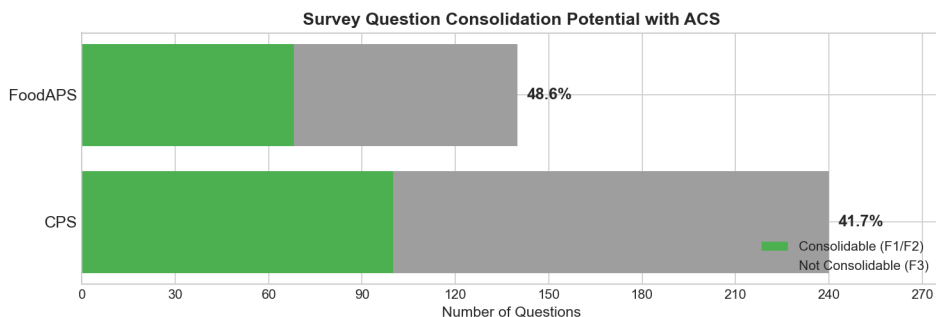
Harmonization Framework

Adopted standard survey harmonization classification:

Level	Definition
F1	Directly consolidable — no transformation needed
F2	Consolidable with transformation — needs recoding
F3	Not consolidable — fundamental barriers

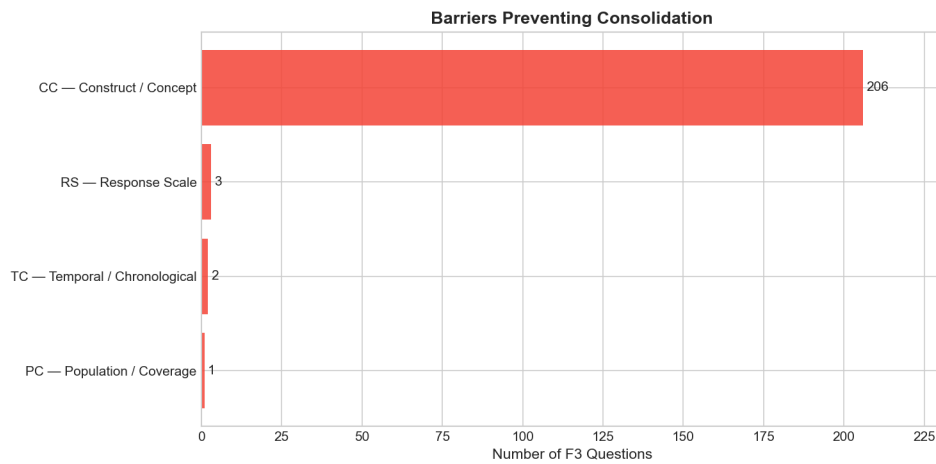
Barrier codes: CC (construct/concept), TC (temporal), RS (response scale)

Headline Results



- **CPS:** 41.7% consolidable (100 of 240 questions)
- **FoodAPS:** 48.6% consolidable (68 of 140 questions)

Why Questions Can't Consolidate

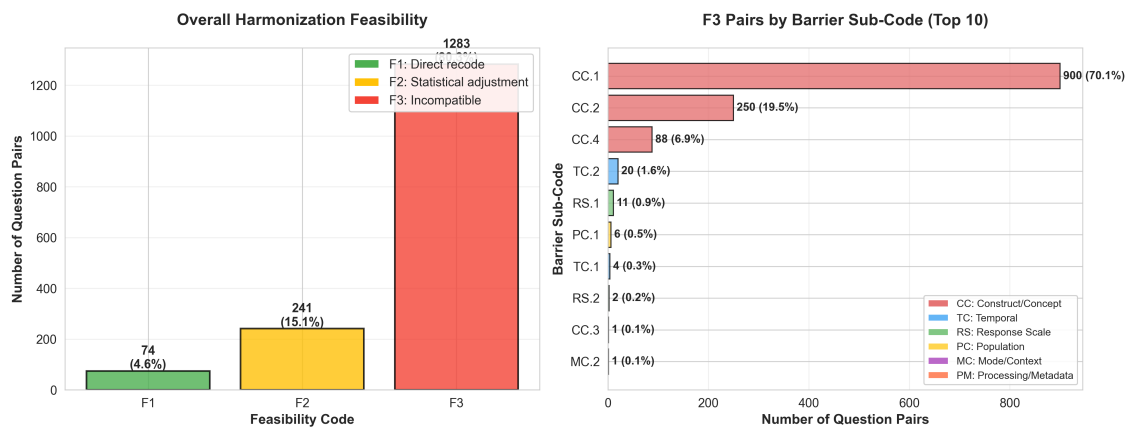


97% of F3 = Construct/Concept differences (CC)

Not fixable barriers — fundamentally different measurements

Harmonization Code Distribution

Harmonization Code Distribution (1,598 Question Pairs)



Left: Overall F1/F2/F3 distribution across 1,598 pairs Right: Top 10 barrier sub-codes for F3 pairs - CC.1 (concept definition) dominates at 70%

Example: High Consolidability (F1)

CPS: “How many hours per week do you USUALLY work?”

ACS: “What is your best estimate of hours per week you usually work?”

- Verdict: **F1** (directly consolidable)
 - Confidence: Borda=1.0, Entropy=1.0
 - Same construct, same framing, same reference
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Example: Medium Consolidability (F2)

CPS: “EXCLUDING overtime, tips, commissions — what is hourly rate?”

ACS: “What is your hourly rate of pay?”

- Verdict: **F2** (consolidable with transformation)
 - Transformation needed: scope alignment (ACS includes all components)
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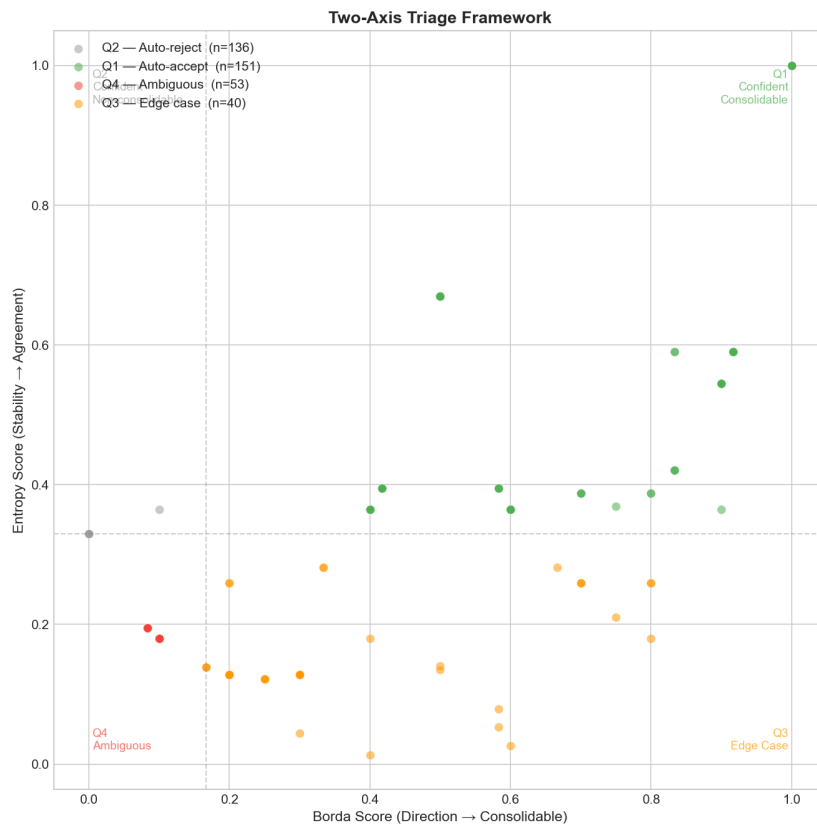
Example: Not Consolidable (F3)

CPS: “Do you WANT to work full time (35+ hours)?”

ACS: “Did this person WORK for pay last week?”

- Verdict: **F3** — Barrier: **CC.1**
 - Same domain (employment), entirely different constructs
 - Preference vs. behavior — not reconcilable
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Expert Review Triage

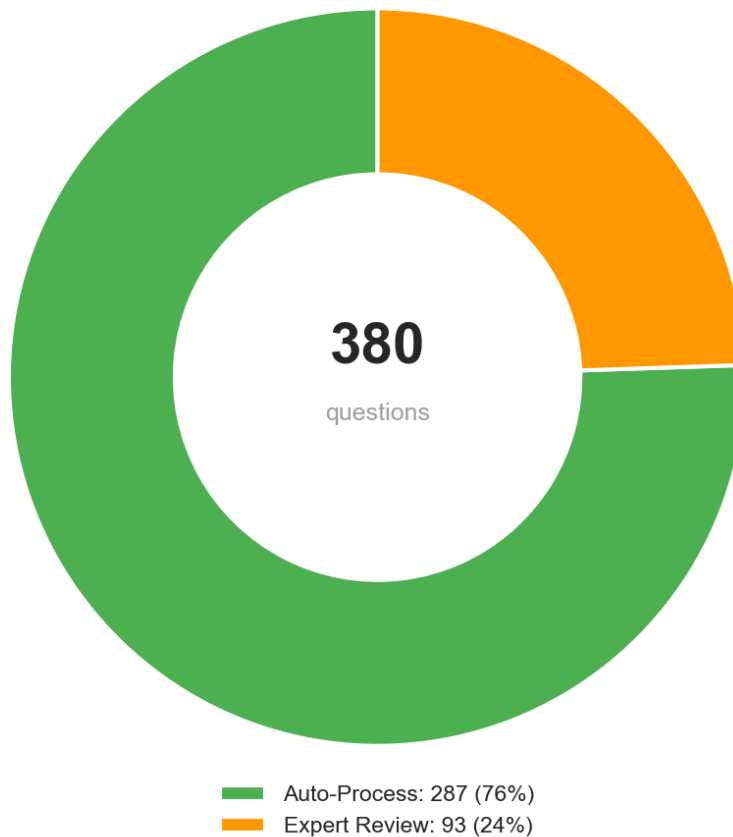


- Two-axis classification: Direction (Borda) × Stability (Entropy)
- **76%** (287 questions) = high confidence → spot-check recommended
- **24%** (93 questions) = flagged for expert review

Traditional: Expert reviews all 1,598 pairs = weeks/months AI-assisted: Expert reviews 93 flagged questions = focused effort on highest-ROI decisions

Expert Review Load

Classification Confidence Distribution



- **This IS the efficiency gain:** Experts review 93 questions, not 1,598 pairs
- AI provides initial judgment, experts provide final validation
- Focus expert time where it adds most value

Deliverables

Primary: Expert review table (`expert_review_combined.csv`) - 380 questions with best ACS matches - F1/F2/F3 classifications with confidence scores - Barrier codes and reasoning for each decision - Triage routing (Q1-Q4) for prioritized review

Breakdown: - 168 consolidable (F1+F2) — ready for validation - 212 non-consolidable (F3) — with documented barriers - 93 flagged for expert review — highest-value decisions - Full methodology — reproducible for additional surveys

What This Proves

- **< 1 day processing time** for work that traditionally takes weeks/months
- **Expert efficiency → faster burden reduction** — identify consolidation opportunities at scale
- **Human oversight preserved throughout** — AI provides initial judgment, experts validate
- **The table is the deliverable** — 380 questions, pre-analyzed, ready for expert review

AI doesn't replace burden reduction goals — it accelerates the path to achieving them through expert efficiency

What's Next

1. Expert validation of recommendations
 2. Expand to additional federal surveys
 3. Application to survey editing/imputation
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Summary

Findings: - ~44% **consolidation potential** — 168 questions could reduce respondent burden - **97% of failures** = construct differences (not fixable)

Method: - **< 1 day processing** for work that traditionally takes weeks/months - **AI provides initial judgment, experts provide final validation** - **The real product:** 380 questions, pre-analyzed, ready for expert review

Impact: Accelerates path to burden reduction through expert efficiency

Expert review table: output/analysis/expert_review_combined.csv Dual value: (1) Identifies burden reduction opportunities (2) Efficiently, preserving expert judgment

For AI methodology validation details, see Report 3B

Questions?

Contact: [your info]

Repository: [link]

Expert review tables: [link to output]